Gut Microbiome Composition and Differential Abundance Across Dietary Groups

Data source: ENA ERP000133
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01 Study Objective and Analytical Questions

Study Objective

Characterize and compare gut microbial communities between two dietary groups using 16S rRNA sequencing data.

Analytical Questions

- What is the overall taxonomic composition?
- Does within-sample microbial diversity differ by diet?
- Does overall community composition differ by diet?
- Which taxa are differentially abundant between groups?

Diet:
Mediterranean (n=15)
Vegetarian (n=14)

→ Gut Microbiome →

Composition
Diversity
Differential Abundance

02. Overall Analytical Workflow

FASTQ files

- Quality profiles.
- Read-length distribution.

Quality assessment and preprocessing

- Filter and trim.
- Learn the error rates.
- Sample inference.

Construct ASV table

Taxonomy Assignment

- RDP release 19 classifier.
- Phylogenetic tree.

Data Integration

Construction of a 'Phyloseq' object.

Main Analysis

- Taxonomic composition.
- Alpha/Beta diversity.
- Differential abundance.

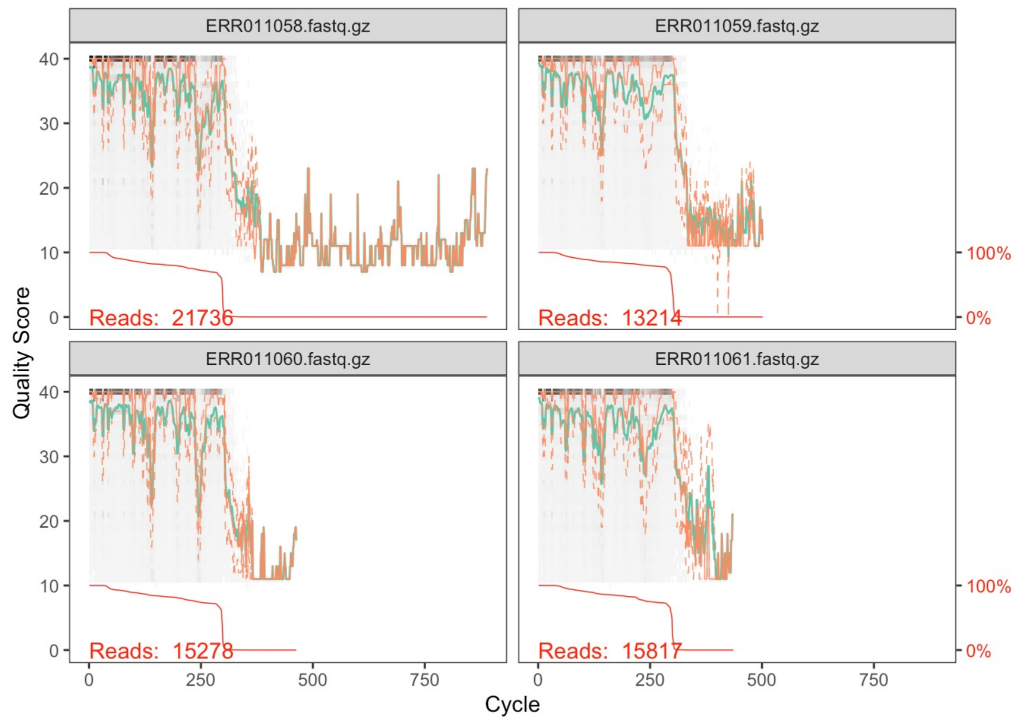
03.

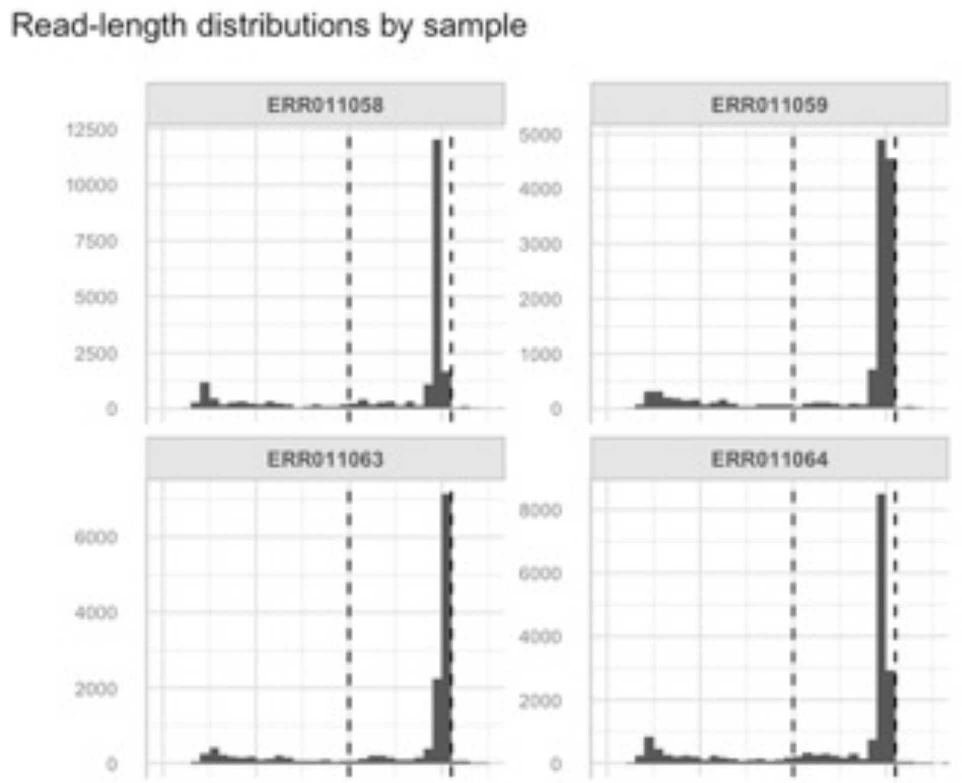
Raw Data and Quality Control

Quality profiles

Input Data

- ENA Project: ERP000133
  29 samples
  16S rRNA amplicon sequencing
  LS454 platform
- Single-end reads





Metrics Examined

- Reads per sample
- Quality-score profiles
- Sequence-length distribution
- Unusual short/long reads

Diet	# Sample	# Read	Mean read-length	Median read-length
Mediterranean	15	243231	262.2639	298
Vegetarian	14	226864	254.2552	298
Overall	29	470095	258.3989	298

Quality filtering and sequence trimming

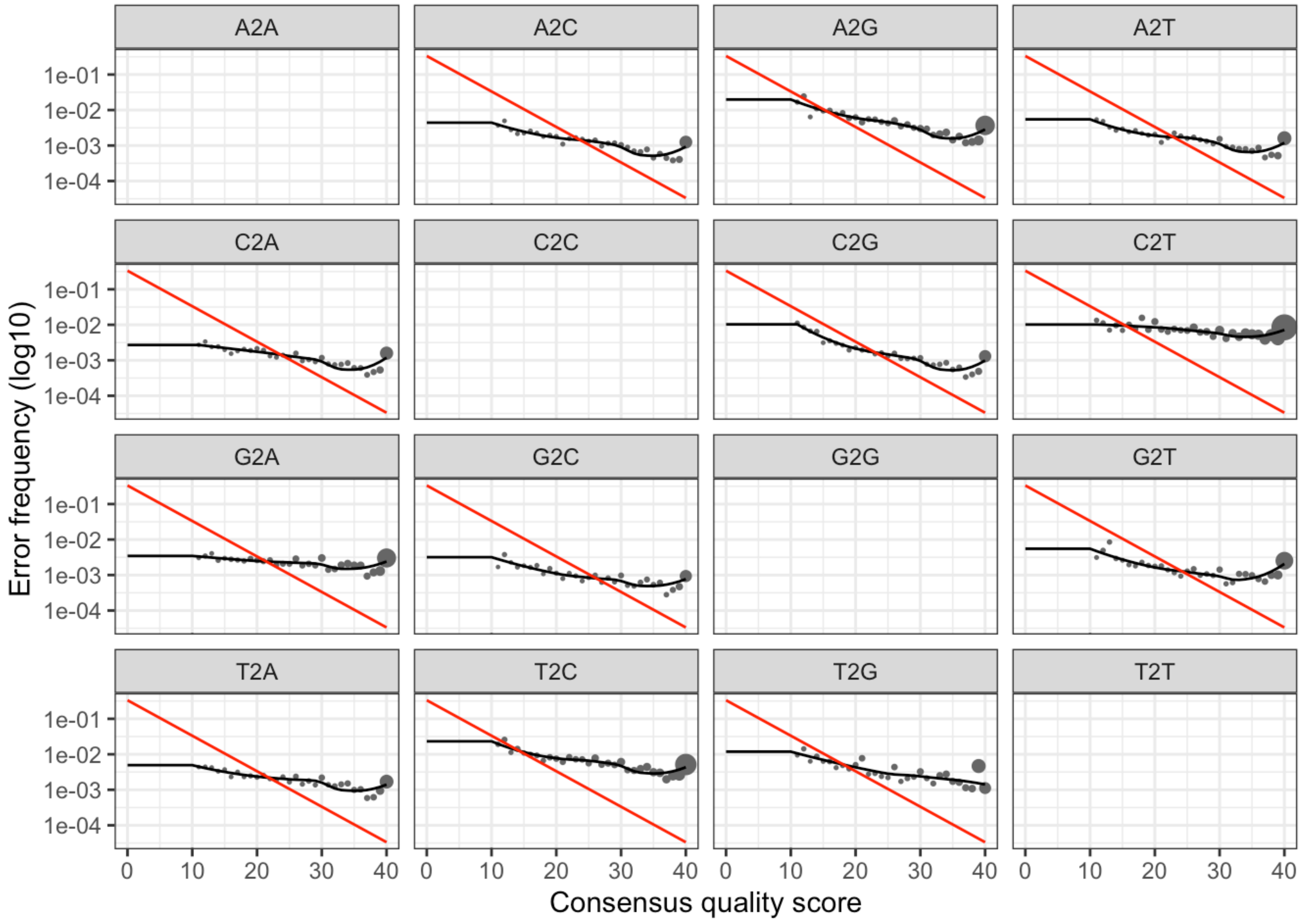
Parameter	Criterion
Maximum length	310 bp (before trimming and truncation)
Minimum length	200 bp (after trimming and truncation)
Ambiguous bases	Remove reads containing Ns
Expected errors	Reads with higher than 2 "expected errors" will be discarded
Truncation quality score cutoff	Truncate reads at the first instance of a quality score less than or equal to 2 (Default)

Retention rate for each dietary group

Diet	# Samples	Mean	Median	Overall	Total reads in	Total reads out
Mediterranean	15	0.786	0.785	0.787	243231	191333
Vegetarian	14	0.752	0.755	0.749	226864	169904

DADA2 workflow

Filtered reads → Error learning → Dereplic-ation → Denoising → Sequence table → Chimera removal → ASVs



ASV table matrix contains 29 rows (samples) and 1716 columns (ASVs).

The retention rate is 80.98%.

Reads throughout the DADA2 pipeline

Diet	Mean reads	Mean filterd	Mean denoised	No chimeras	No chimera rate
Mediterranean	16215.40	12755.53	12422.67	10246.40	0.63
Vegetarian	16204.57	12136.00	11565.93	9167.43	0.57

Taxonomic Assignment and Phylogenetic Tree Construction

Taxonomic assignment

- ASV sequences → RDP reference database
- Taxonomic levels:
  - Kingdom
  - Phylum
  - Class
  - Order
  - Family
  - Genus

Phylogenetic tree

ASV sequences

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Multiple sequence alignment

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Neighbor-Joining starting tree

↓

Maximum-likelihood optimization
(GTR + Γ + I)

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Main Analysis

Data Integration

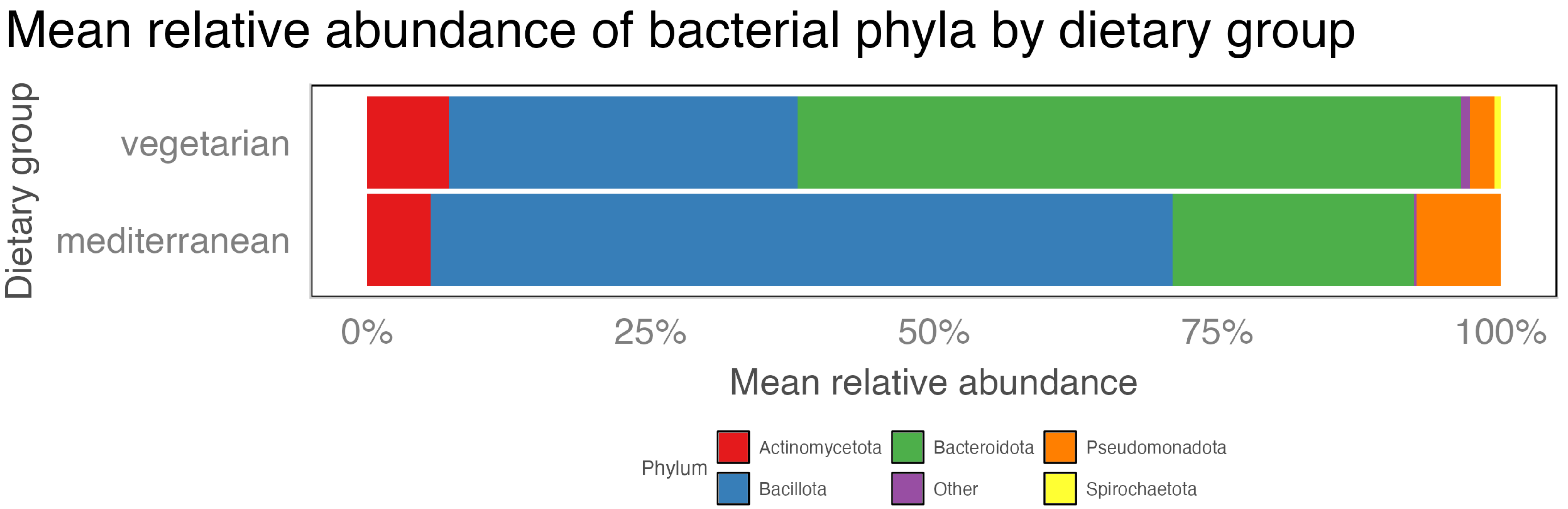
ASV abundance table
+
Taxonomy data
+
Participant table
+
Phylogenetic tree
↓
PHYLOSEQ
↓
Additional filtering
↓
Analysis-ready dataset

Additional taxonomic filtering

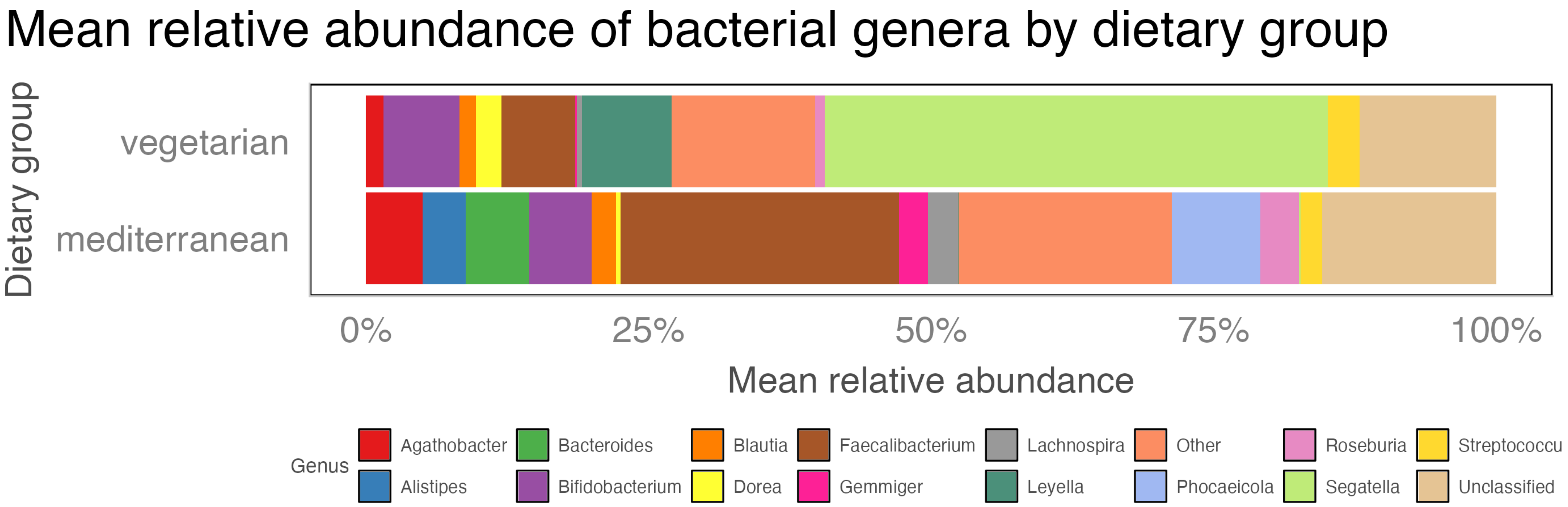
- Retain Bacteria (removed 2 ASVs without Kingdom classification.
- No *Chloroplast* or *Mitochondrial* sequences exist.
- The minimum abundance is 2.
- 1714 ASVs.

Taxonomic composition by dietary group

Phylum-level



Genus-level



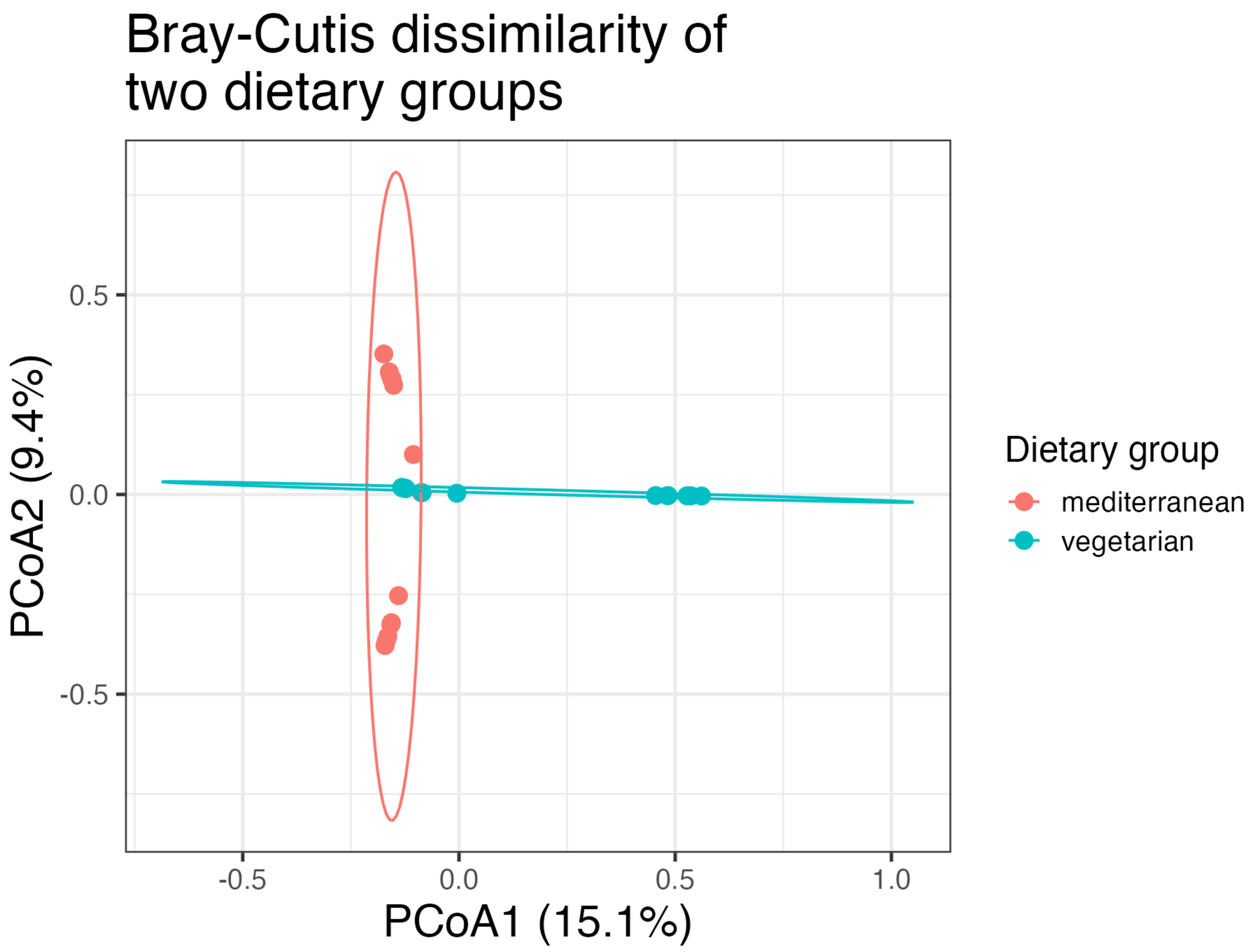
Alpha diversity

Mediterranean group had higher within-sample diversity than Vegetarian group.

Metrics	p-value	Adjusted p-value
Observed	0.025	0.025
Shannon	0.002	0.003
Simpson	0.001	0.003

Beta diversity

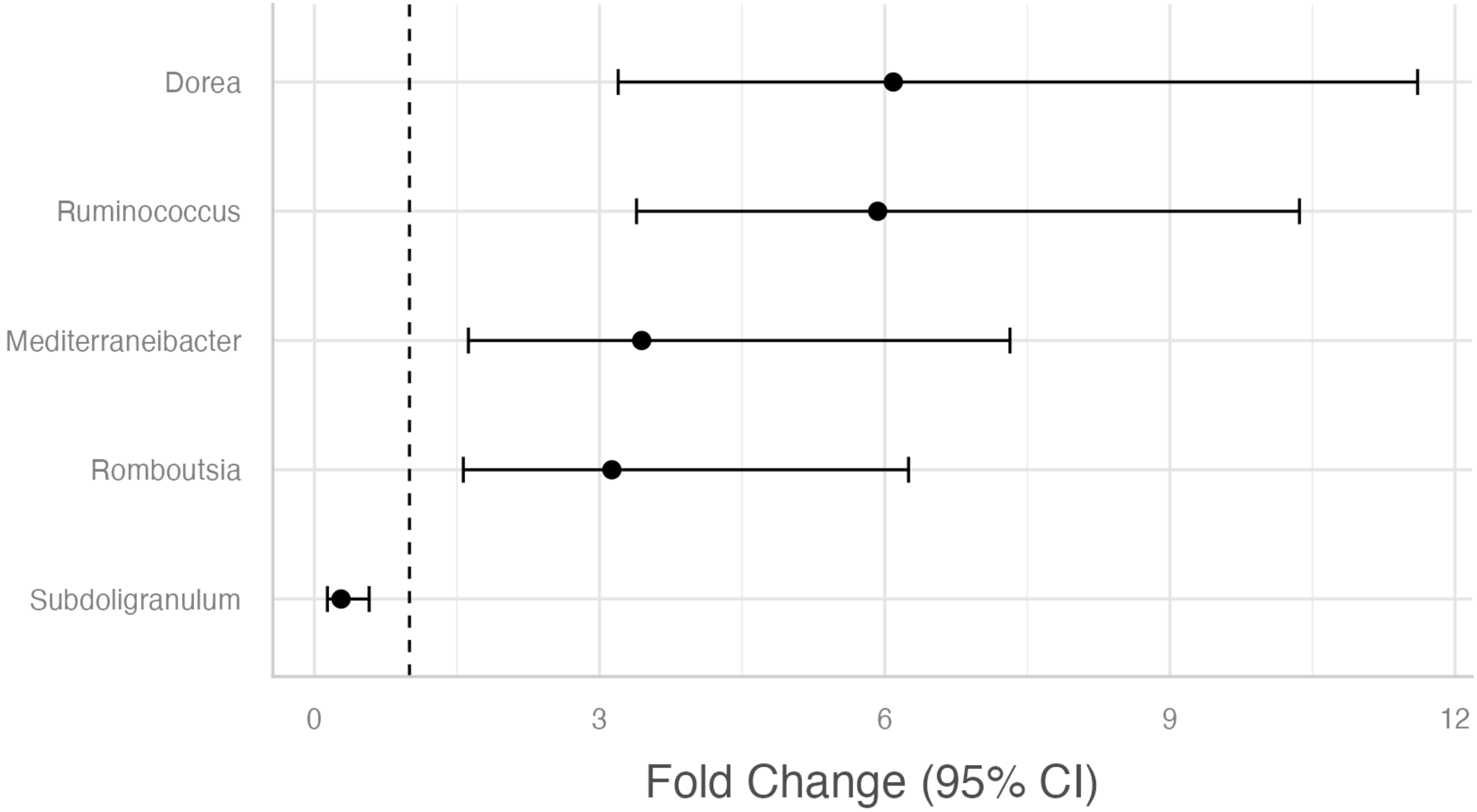
Beta diversity reveals differences in overall community structure.



	R^2	p-value
PERMANOVA	0.099	0.001
Dispersion	-	0.305

Differential Abundance Analysis with ANCOM-BC2

Differentially abundant genera by diet



Taxon	Fold change	95% CI
Dorea	6.09	(3.20 – 11.61)
Ruminococcus	5.93	(3.39 – 10.36)
Mediterraneibacter	3.44	(1.62 – 7.32)
Subdoligranulum	0.28	(0.14 – 0.58)
Romboutsia	3.13	(1.57 – 6.25)

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Key findings

Within-sample diversity

Dietary groups differed in microbial diversity.

Overall community structure

Diet explained approximately 10% of between-sample variation.

Specific taxa

ANCOM-BC2 identified taxa associated with dietary group.

Biological interpretation

Long-term diet may shape the gut microbiome through differences in fermentable carbohydrates, dietary fiber, fat, and protein intake.

Limitations, Potential Improvements, and Challenges

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Limitation

- Small sample size.
- Potential confounders.
- Dated sequencing method.
- Genus-level classification.

Potential improvement

- Including relevant explanatory variables.
- Higher level taxonomy.
- Validate findings.

Challenges

- Maintain sequence identifiers consistency.

thank you!

Relevant documents can be found at:

https://lilyzhang4499-cmd.github.io

Tools

Steps	Tools
Filtering, Error learning, ASV inference	DADA2
Taxonomy assignment	RDP 19
Alignment	DECIPHER
Phylogenetic tree	phangorn
Data integration	phyloseq
Alpha diversity	phyloseq
Beta diversity	vegan
PERMANOVA	vegan
Dispersion	vegan
Differential abundance	ANCOMBC
Visualization	ggplot2

Framework

Within group diversity

- Diversity metrics
- Non-parametric approach

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Between group composition

- Bray-Curtis dissimilarity
- PERMANOVA
- Check multivariate dispersion

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Specific taxa

- ANCOM-BC2
- FDR correction

Methods selection

Methods	Rational
Shannon index	Captures both richness and evenness of taxonomic abundance.
Simpson index	Another measure of diversity to put more weights on the dominant taxa.
Bray-Curtis dissimilarity	Quantifies differences in community composition using taxon relative abundances.
PERMANOVA	Formally tests whether overall microbial community composition differs between dietary groups.
ANCOM-BC2	Identifies differentially abundant taxa while accounting for the bias and multiple testing.